Assignment 2

You are given a 3D domain, owned by px X py X pz processes. Each sub-domain size is nx X ny X nz. Iterate over T steps, Dimension of Data[F] per process = nx * ny * nz doubles, F is the number of fields/variables.

Communication: halo exchange for d-point stencil
Computation: d-point stencil (d is input, d < nx, d < ny, d < nz, example values of d are 7, 13, …)
For e.g., average of E,W,N,S,F,B and itself for d=7, i.e. value (P, t+1) = [value (P_W, t) + value (P_E, t) + value (P_N, t) + value (P_S, t) + value (P_F, t) + value (P_B, t) + value (P, t)] / 7
Each process computes the count of isovalues (i.e. the count of points in the domain that would be equal to a given isovalue, refer to Lectures 17 and 18)

Root process calculates the total count of isovalues for every field across all time steps, i.e. there will be T values for field #1, T values for field #2, …. , and T values for field #F.

The initial data must be randomly generated using the value "seed" as shown below.

```
srand(seed);
for (int i=0; i<F; i++)
  for (int j=0; j<arrSize; j++)            // arrSize = nx * ny * nz
    data[i][j] = (double)rand()*(myrank+1)/(110426.0+i+j);
   // you are free to use any data layout but the initialization should be done like this.
```

The input to the program is d, ppn, px, py, pz, nx, ny, nz, T, random seed, F, isovalue

The output (by rank 0) should be **T+1 lines**, the first T lines contain F numbers, followed by the time:
<count_field1> … <count_fieldF> // for time step 1
…
<count_field1> … <count_fieldF> // for time step T
<time> [put a new line at the end]

Final configurations for the report: (there are a total of 8 configurations)

for execution in 1 to 5 // repeat each configuration 5 times
P (number of processes) in 32, 48, 64, 96 (4 configurations)

if P = 32 then px = 4, py = 4, pz = 2, ppn = 32
if P = 48 then px = 6, py = 4, pz = 2, ppn = 48
if P = 64 then px = 4, py = 4, pz = 4, ppn = 32
if P = 96 then px = 6, py = 4, pz = 4, ppn = 48

nx = ny = nz = 120, nx = ny = nz = 240 (2 configurations)
d = 7
ppn = as per above // For PARAM Rudra (note: this will be set as per the test environment)
T = 5
seed = 1000

F = 2
isovalue = 500

You **must only** use the MPI functions discussed in class. You may optimize your code using concepts learnt in class. Submit your optimized code only.

Plot the time (in seconds) for each data size per process count. Use boxplots (from the 5 executions) for every data point in a plot. Submit one plot file only. Time in seconds on the y-axis and processes (P) on the x-axis.